Baohua Yang

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PROFILE

- Systems architect and technical leader with nearly 20 years of turning innovations into production systems, from 10 Gbps network processing and blockchain platforms to agentic AI.
- Designs agentic AI platforms end to end: multi-agent orchestration, MCP tool integration, context management, evaluation, and runtime safety; hands-on in Python and Go.
- Three-term Technical Steering Committee member of Hyperledger (now Linux Foundation Decentralized Trust); 10 granted patents, 20+ papers, and technical books with 100M+ reads.

TECHNICAL SKILLS

- **Distributed Systems:** high performance, consensus, data-center, monitoring and failure diagnosis, Kubernetes, Go, Python, C/C++.
- **Agentic AI:** multi-agent orchestration (supervisor pattern), MCP, RAG, context and memory management, evaluation and telemetry, LLM inference.
- **Security:** prompt-injection and jailbreak defense (NVIDIA NeMo Guardrails), OPA policy-as-code authorization, PKI and X.509 identity, network security.

EXPERIENCE

Principal Architect, *Oracle* **Aug 2017 – Present**

- Own architecture for Oracle Blockchain Platform and Oracle TSAM, serving Fortune 500 customers (~$200M ARR, ~50 engineers); partner with the Database team.
- Architected and shipped an AI agent that answers questions over enterprise data and diagnoses failures; raised accuracy from ~30% to 90%+ and cut latency by 50%.
- Built the agent's safety layer: NeMo Guardrails input, output, retrieval and tool rails against direct and indirect prompt injection, jailbreaks and data leakage; OPA policy for scoped tool permissions.
- Designed a supervisor-based multi-agent framework (similar to LangGraph), replacing fixed call chains with dynamic runtime orchestration; drove the adoption in DB's AI agent feature.
- Influenced the Oracle DB AI roadmap (RAG, context management); selected for Oracle's AI Champion program (~100 company-wide).
- Architected three generations of Oracle Blockchain Platform (VMs, containers, Kubernetes); cut provisioning from minutes to seconds and cost by 50%+; scaled the platform to 10,000 TPS and several hundred tenants.
- Led resolution of the hardest production failures, e.g. a consensus freeze (fail-closed fix shipped upstream) and state-database corruption (10K+ TPS Go ledger parser).

Research Scientist, *IBM Research* **Jan 2013 – Aug 2017**

- Early designer and core developer of Hyperledger Fabric, one of the first enterprise blockchain platforms; led the applied research behind it.
- Architected IBM's enterprise blockchain platforms in Go (X.509 identity, CA hierarchy, revocation, endorsement policy), sustaining 1,000+ TPS per cluster across hundreds of production clusters.

OPEN SOURCE & COMMUNITY

Hyperledger Foundation TSC Member (now LF Decentralized Trust) **2017–2019; 2020–2021**

- Founding developer; shaped technical strategy and governance across Hyperledger over three terms and contributed to ISO/TC 307 blockchain standardization.
- Maintainer of Hyperledger projects; open-source community mentor and technical writer.

EDUCATION

Ph.D. in Control Science and Engineering, ***Tsinghua University*** **Sep 2007 – Jan 2013**

Research: high-performance networking and security; built a 10 Gbps UTM system.

IBM Ph.D. Fellowship Award (9 winners, 2011).

Visiting Scholar, ***EECS, University of California, Berkeley*** **Sep 2010 – Sep 2011**

Research: software-defined and data-center networking.

Bachelor of Science and Engineering, ***Tsinghua University*** **Aug 2003 – Jul 2007**

PATENTS

- 10 granted patents in distributed ledgers, cloud, and networking, including US 11,822,538, US 11,741,075, US 11,546,425, and US 10,983,816.
- 3 AI patent filings (2024–2026): long-context management for agentic systems, enhanced NL2SQL querying, and AI knowledge-base protection.

SELECTED PUBLICATIONS

- LazyCtrl: A Scalable Hybrid Network Control Plane Design for Cloud Data Centers. K. Zheng, L. Wang, B. Yang, Y. Sun, S. Uhlig. IEEE Transactions on Parallel and Distributed Systems, 28(1):115–127, 2017.
- Keep Forwarding: Towards K-link Failure Resilient Routing. B. Yang, J. Liu, S. Shenker, J. Li, K. Zheng. IEEE INFOCOM 2014.
- Practical Multi-tuple Packet Classification using Dynamic Discrete Bit Selection. B. Yang, J. Fong, W. Jiang, Y. Xue, J. Li. IEEE Transactions on Computers, 63(2):424–434, 2014.
- Data-Driven Network Connectivity. J. Liu, B. Yang, S. Shenker, M. Schapira. ACM HotNets-X, 2011.

20+ peer-reviewed papers in total; full list on Google Scholar.

SELECTED BOOKS AND TECHNICAL GUIDES

- Blockchain: Principle, Design and Applications (2017, 2 editions), China Machine Press.
- Docker Container Technology in Action (2015, 4 editions), China Machine Press.
- Open-source books Docker Practice (26K+ GitHub stars) and Blockchain Guide (7K+), 100M+ reads; 4,800+ GitHub followers.
- Open-source AI guides (2026): Agentic AI, Harness Engineering, AI Security, Context Engineering, LLM Internals.